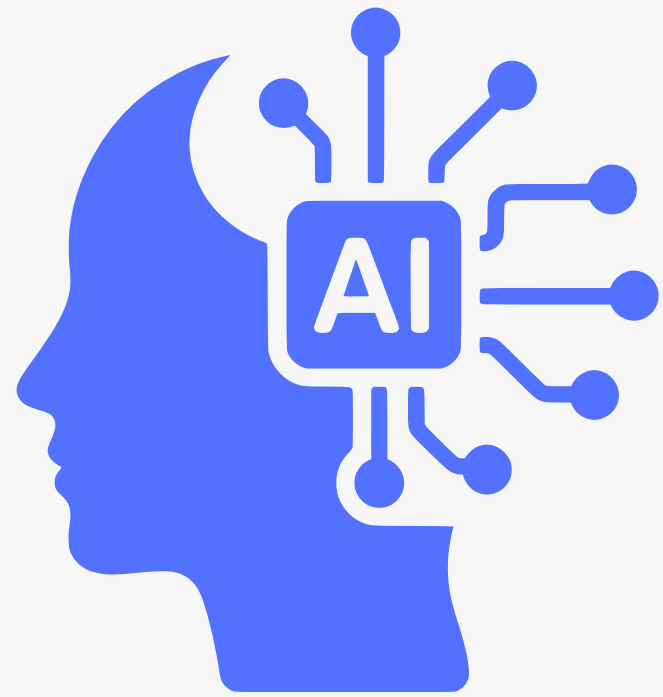




MULTIMODAL API

WHAT IF AI COULD NOT ONLY READ YOUR TEXT, BUT ALSO SEE IMAGES, UNDERSTAND DOCUMENTS, LISTEN TO YOUR VOICE, AND SPEAK BACK TO YOU?





WHAT IS AN API?

- API = Application Programming Interface
- Allows two applications to communicate
- Sends requests and receives responses
- Connects your app with external services
- Client App → API Request → AI Model → API Response
→ Client App
- OpenAI API, Gemini API, Anthropic Claude API, Groq API,
DeepSeek API





WHAT IS A MULTIMODAL API?

- Processes multiple types of data
- Works with text, images, audio, and documents
- Understands different input formats together
- Enables more human-like interactions
- Image / Audio / Text → Multimodal API → AI Processing → Response





VISION APIS

- Allow AI to analyze images
- Detect objects, scenes, and visual content
- Extract useful information from images
- Understand what is happening in a picture
- Image → Vision API → Image Analysis → AI Response
- OpenAI Vision, Gemini Vision, Azure Computer Vision





OCR (OPTICAL CHARACTER RECOGNITION)

- Extracts text from images
- Converts scanned documents into editable text
- Reads printed and handwritten content
- Commonly used in document processing
- Document/Image → OCR API → Extract Text → Process Text
- Google Vision OCR, Azure OCR, AWS Textract, Gemini OCR





STT (SPEECH-TO-TEXT)

- Converts speech into text
- Helps AI understand spoken language
- Supports voice commands and transcription
- Voice Input → STT API → Text Output
- OpenAI Whisper, Google Speech-to-Text, Azure Speech Services





TTS (TEXT-TO-SPEECH)

- Converts text into natural speech
- Creates AI-generated voices
- Supports multiple languages and accents
- Text → TTS API → Audio Output
- OpenAI TTS, ElevenLabs, Google Text-to-Speech, Azure Speech





IMAGE UNDERSTANDING

- Goes beyond object detection
- Understands image meaning and context
- Can answer questions about images
- Enables visual reasoning
- Image + Question → AI Model → Context
Understanding → Response
- GPT-4o Vision, Gemini Vision, Claude Vision





VOICE PIPELINES

- Combine STT + LLM + TTS
- Enable real-time voice conversations
- Foundation of AI voice assistants
- Creates natural human-AI interaction
- User Voice → STT (Speech-to-Text) → LLM Processing → TTS (Text-to-Speech) → AI Voice Response
- OpenAI Realtime API, Gemini Live API, ElevenLabs, Deepgram

